

FUAD AHMED

Full-Stack Software Engineer & Computer Systems Researcher

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ACADEMIC & RESEARCH PROFILE

Results-driven Computer Science graduate and Systems Engineer with a **3.20 Major GPA** and proven experience designing high-concurrency backend architectures, reactive cross-platform mobile systems, and geospatial engines. Combines rigorous computational foundations (distributed systems, database internals, and concurrency algorithms) with practical engineering leadership in multi-stakeholder and operational environments. Seeking admission to a **Master of Science (M1/M2) in Computer Science / Software Architecture & Distributed Systems** in France, with eligibility for the **France Excellence Eiffel Scholarship**.

EDUCATION & ACADEMIC ACHIEVEMENTS

Bachelor of Science in Computer Science

St. Mary's University, Addis Ababa, Ethiopia

Graduated: July 2022 | Medium of Instruction: 100% English

- **Major GPA: 3.20 / 4.00 | Cumulative GPA: 3.04 / 4.00**
- **Core Computational Coursework:**
- Data Structures & Algorithms (Advanced Algorithmic Complexity, Graph Algorithms, Sorting & Searching)
- Database Systems & Architecture (Relational Normalization, Indexing, B-Trees, Query Optimization, ACID Properties)
- Operating Systems & Systems Programming (Process Scheduling, Thread Synchronization, Memory Hierarchy, IPC)
- Computer Networks & Distributed Communication (TCP/IP Protocols, Sockets, Network Topology, Security)
- Object-Oriented Analysis & Design (Design Patterns, SOLID Principles, UML Architectural Modeling)
- Software Engineering & Quality Assurance (Testing Methodologies, SDLC, CI/CD Fundamentals)
- **BSc Capstone Project:** Multi-tiered web and database management system with role-based access control (RBAC), transactional consistency, and normalized MySQL schemas.

Executive Certification: Capital Markets & Securities Analysis

Addis Ababa University, School of Commerce (in collaboration with Ethiopian Capital Market Authority)

Completed: September 2024 (50-hour intensive certification)

- **Focus Areas:** Financial market microstructure, equity valuation, clearing and settlement mechanisms, risk assessment models, and transactional integrity.

SOFTWARE SYSTEMS & PRODUCTION PROJECTS

Addis Condo Finder — Geospatial Distributed Property & Mapping Platform

Architecture: Flutter (Dart), Laravel 11/13 REST API, PostgreSQL/MySQL, Filament Admin, OpenStreetMap

- **Spatial Query Optimization:** Designed bounding-box spatial indexing algorithms

(`bbox=minLat,minLng,maxLat,maxLng`) to serve viewport-constrained geospatial payloads, eliminating server memory overhead.

- **Token-Matching Search Engine:** Engineered an any-order multi-token search pipeline indexing condominium names, project IDs, and block numbers with sub-50ms query response times.
- **Consensus Verification & Security:** Implemented a multi-user crowdsourced verification engine with client write rate-limiting (20 requests/minute per user) and atomic database unique-constraint handling to prevent verification races.

****LuckyDraw Engine — High-Concurrency Financial Gaming & Ledger Architecture****

Architecture: Laravel 13, Livewire 4, MySQL (InnoDB), Telegram Mini App API, REST Webhooks

- **Atomic Concurrency Guards:** Implemented strict pessimistic row-level locking (`lockForUpdate`) on draw execution and prize distribution, mathematically guaranteeing zero double-payouts and single-draw dispatch under concurrent client traffic.
- **Integer-Cent Exact Accounting:** Formatted all currency balances and ticket prices into integer cents (`tests/Unit/MoneyTest.php`) to eradicate floating-point rounding errors across multi-tiered jackpot distributions.
- **Fail-Closed Payment Verification:** Integrated automated payment-verification webhooks for mobile money providers (Telebirr/CBE) with unique reference deduplication to prevent replay attacks.
- **Automated Testing Suite:** Developed a suite of 25+ automated feature, unit, and concurrency tests validated directly on production-grade MySQL engines.

****Money Manager — Offline-First Reactive Personal Finance System****

Architecture: Flutter, Dart, Riverpod 2.0, SQLite (sqflite), Material 3 Design System

- **In-Memory Cache & Synchronous Reads:** Structured an `AppRepository` abstraction maintaining an in-memory state cache for instant UI rendering while persisting incremental row mutations asynchronously to SQLite.
- **Security & Biometrics:** Engineered biometric fingerprint/PIN authentication with failed-attempt lockout security and OS-level `FLAG_SECURE` window protections preventing app switcher data leakage.
- **Automated Scheduling:** Built background local notification schedulers and auto-posting engines for recurring debts, upcoming bills, and savings goal contributions.



PROFESSIONAL EXPERIENCE

****Full-Stack Software Engineer (Independent & Contract)****

Addis Ababa, Ethiopia | Nov 2023 – Present

- Architected, tested, and deployed end-to-end web and mobile applications using Flutter, Laravel, and relational database systems.
- Designed resilient Android background processing pipelines, leveraging Foreground Services, WorkManager, and battery-optimization bypass mechanisms for high-reliability field execution.
- Authored clean, maintainable documentation, schema migrations, and CI workflows across multiple open-source and proprietary codebases.

****Web Developer & UI/UX Systems Specialist****

Aduutech, Addis Ababa, Ethiopia | Feb 2023 – Nov 2023

- Developed responsive enterprise web platforms and internal inventory management systems utilizing PHP, Laravel, and MySQL.
- Re-engineered inventory tracking workflows handling 1,000+ daily operational records, reducing order processing latency by 35%.

- Actively participated in weekly sprint planning, rigorous peer code reviews, and schema optimization sessions.

****Operations & Systems Lead (Hospitality Management)****

Addis Ababa, Ethiopia | Jan 2022 – Jan 2023

- Supervised end-to-end property operations, front-desk hospitality management, guest reservations, and supplier logistics.
- Implemented digital POS and computerized booking systems, streamlining billing transparency and financial record-keeping.
- Demonstrated high leadership maturity, managing cross-functional service teams, resolving high-pressure client escalations, and optimizing resource allocation.

****Database Design Intern****

Ministry of Revenues, Addis Ababa, Ethiopia | Oct 2021 – Nov 2021

- Designed normalized 3NF relational schemas for taxpayer agent registry and auditing modules.
- Collaborated with senior systems architects to write indexing strategies, foreign key constraints, and audit logging triggers.

TECHNICAL COMPETENCIES

- **Programming Languages:** PHP (8.x), Dart, JavaScript (ES6+), Python, SQL, C++, Java/Kotlin basics
- **Frameworks & Libraries:** Laravel, Flutter, Riverpod, Livewire, Filament, Vue.js / React basics, Bootstrap, TailwindCSS
- **Databases & Storage:** MySQL, PostgreSQL, SQLite, Redis caching fundamentals
- **Architecture & Patterns:** Microservices & Modular Monoliths, RESTful API Design, MVC, Repository Pattern, Offline-First Architecture, Concurrency Control (Row Locking, ACID)
- **DevOps & Developer Tools:** Git, GitHub, Linux (Ubuntu/Debian, Bash scripting), Docker fundamentals, Apache/Nginx, cPanel
- **Testing & Quality Assurance:** PHPUnit, Pest PHP, Concurrency Test Suites, Flutter Test Harness

LANGUAGES & INTERCULTURAL COMPETENCIES

- **English:** Fluent (Full Professional & Academic Proficiency — Official Medium of Instruction Certified)
- **Amharic:** Native
- **Oromiffa:** Professional Working Proficiency
- **French:** Beginner / Elementary (A1 in-progress — actively acquiring French language and culture)
- **Intercultural Aptitude:** Proven capacity to collaborate in diverse, multi-lingual teams, high adaptability, and strong work ethic in rigorous academic environments.

VERIFIABLE CREDENTIALS & REFERENCES

- **GitHub Repository:** <https://github.com/fuad745>
- **Master Scholarship Dossier:** <https://github.com/fuad745/scholarship-dossier-2026>
- Academic transcripts, official English Medium of Instruction certificate, and faculty recommendation letters from St. Mary's University available immediately.